FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2021

TEST DATE: 18-04-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $\frac{P_P}{P_Q}=\left(\frac{T_Q}{T_P}\right)^2\left(\frac{A_P}{A_Q}\right)^2=\left(\frac{4}{6}\right)^2\times\left(\frac{2}{1}\right)^2=\frac{16}{9}$

2. C

Sol. $g\sin\theta\cos\theta$

θ

$g\sin\theta$ $g\sin^2\theta$

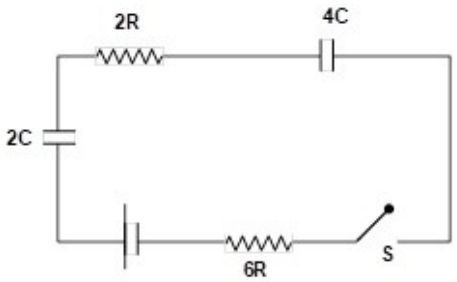
θ

3. C

Sol. $R_{eq}=8R$

$C_{eq}=\frac{4C}{3}$

$\tau=R_{eq}\times C_{eq}=\frac{32CR}{3}$



4. B

Sol. $mv_0\frac{\ell}{2}=4m\left(\frac{\ell}{\sqrt{2}}\right)^2.\omega$

$mv_0\frac{\ell}{2}=4m\frac{\ell^2}{2}.\omega$

$\omega=\frac{v_0}{4\ell}$

5. C

Sol. $v_1=\left(\frac{m_1-em_2}{m_1+m_2}\right)v_0$

$m_1>em_2\Rightarrow\frac{m_1}{m_2}>e$

6. A

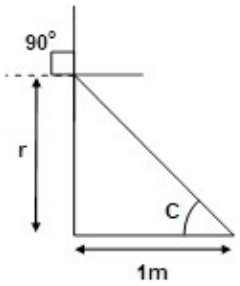
Sol. $\sin C=\frac{1}{\mu}=\frac{3}{5}$

$\tan C=\frac{r}{1}$

$r=\tan C=\frac{3}{4}\ m$

$(1-kt)=\frac{3}{4}$

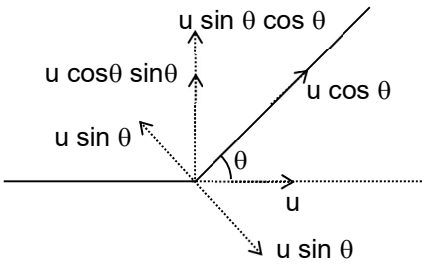
$\frac{1}{8}\times t=\frac{1}{4}\Rightarrow t=2\,sec$



7. B

Sol. $v_y = 2u \sin\theta \cos\theta$

$= u\sin 2\theta$



8. D

Sol. P = kt

FV = kt $\Rightarrow\ m\frac{dv}{dt}.v=kt$

$m\int_0^v vdv=k\int_0^t tdt$

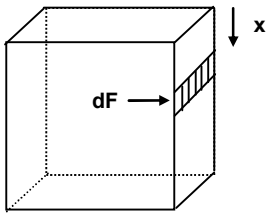
$\frac{v^2}{2}\propto\frac{t^2}{2}$

$S\propto t^2$

9. B

Sol. $P = \int_0^x \rho_0 g\left(1+\frac{x}{h}\right)dx$

$P = \rho_0 g\left(x+\frac{x^2}{2h}\right)$



Force on the face ABCD

$F = \int_0^h \rho_0 g\left(x+\frac{x^2}{2h}\right)bdx$

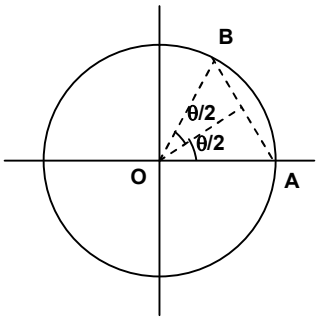
$F = \frac{2}{3}\rho_0 bgh^2$

10. B

Sol. $\theta = \frac{1}{2}\alpha t^2$

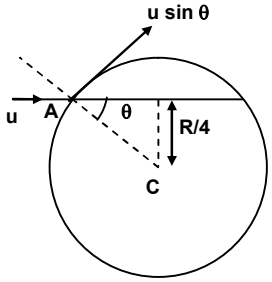
$AB = 2R\sin\frac{\theta}{2}$

Average velocity $= \frac{AB}{t} = \frac{2R}{t}\sin\left(\frac{\alpha t^2}{4}\right)$



11. A

Sol. $u\sin\theta = \left(\frac{u}{4}\right)$



12. B

Sol. PT^{-5} = constant

$PV^{5/4}$ = constant

$\alpha = \frac{5}{4}$

$C = C_V + \frac{R}{(1-\alpha)}$

$C = \frac{3R}{2} + \frac{R}{1-\frac{5}{4}} = \frac{3R}{2} + \frac{4R}{(-1)}$

$C = \frac{3R}{2} - 4R = \frac{3R-8R}{2} = \frac{-5R}{2}$

13. D

Sol. Work done = $q(V_C - V_A) = \frac{q\lambda}{2\pi\varepsilon_0}\ln\left(\frac{a+4R}{a}\right)$

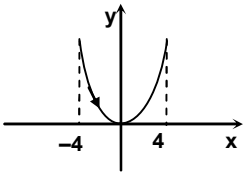
14. A

Sol. $F = iB.\ell = 4 \times 0.04 \times 8$

F = 1.28 newton

ma = 1.28

$a = \frac{1.28}{0.2} = \frac{12.8}{2} = 6.4\ m/s^2$



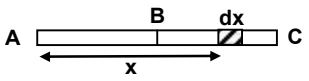
15. A

Sol. $di = \frac{\lambda dx}{\frac{2\pi}{\omega}} = \frac{\omega\lambda dx}{2\pi}$

$dB = \frac{\mu_0 di}{2x}$

$dB = \frac{\mu_0}{2x}.\frac{\omega\lambda}{2\pi}dx$

$\int dB = \frac{\mu_0\omega\lambda}{4\pi}\int_{\ell/2}^{\ell}\frac{dx}{x} = \frac{\mu_0\omega\lambda}{4\pi}\ln 2$

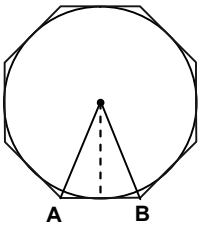


16. D

Sol. Complete hexagon

Total emf = $\pi R^2\alpha$

emf in AB = $\frac{\pi R^2\alpha}{6}$



17. D

Sol. Conceptual

18. B

Sol. Force due to light = $(1+r)\frac{IA}{C} = \frac{1.2I\pi R^2}{C}$

19. A

Sol. $MV = \frac{h}{\lambda}$

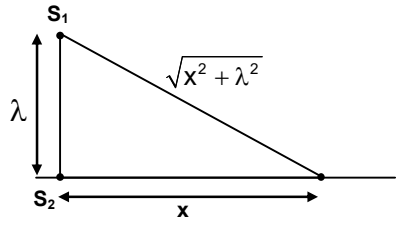
$\frac{1}{\lambda} = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$

20. C

Sol. $\sqrt{x^2+\lambda^2}-x=\left(\frac{\lambda}{2}\right)$

$x^2+\lambda^2=\left(x+\frac{\lambda}{2}\right)^2$

$x=\frac{3\lambda}{4}$



SECTION – B

21. 00100.00

Sol. $N = N_0 e^{-\lambda t}$, $q = q_0 e^{-t/RC}$

$\frac{N}{q}$ is constant when

$\lambda=\frac{1}{CR}$

$R=\frac{1}{\lambda C}=\frac{1}{40\times10^{-6}}\times4\times10^{-3}=100\Omega$

22. 00360.00

Sol. $I=\frac{V}{Z}$

$V_C=IX_C=360$

23. 00063.69

Sol. β =10 log $\frac{I}{I_0}$

$I=10^{-6}$ w/m^2

$P = I.\ 4\pi r^2 = 3.14$ watt

$T=\frac{1\times10^3\times\frac{20}{100}}{3.14}=\frac{200}{3.14}=63.69\,\text{sec}$

24. 00023.70

Sol. Output = 24 – 0.3 = 23.7 V

25. 00012.50

Sol. $I=\frac{I_0}{2}\left(\frac{1}{4}\right)\Rightarrow\frac{I}{I_0}=\frac{1}{8}$

or $\frac{100}{8}\%=12.5\%$

26. 00015.73

Sol. $B=\frac{\mu_0}{4\pi}\times\frac{M}{r^3}$

$M=15.73\times10^{22}$ Am^2

27. 04485.71

Sol. $F = -mg \sin(\theta + \phi)$

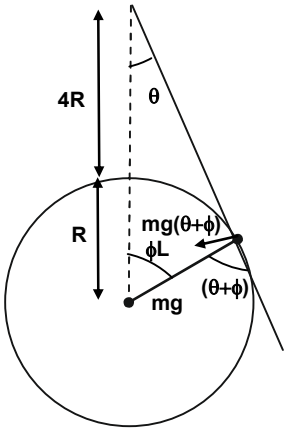
$ma = -mg(\theta + \phi)$

$a = -g\left(\frac{x}{4R} + \frac{x}{R}\right)$

$\Rightarrow a = -\left(\frac{5}{4}\frac{g}{R}\right)x$

….(i)

$T = 4485.71$ sec



28. 00013.25

Sol. $-10.5 = \frac{1}{2}2 \times (4)^2 + PE$

PE = –26.5 J

Potential = -13.25 J/kg

29. 00050.00

Sol. $F = 10\ g \sin 30 = 10 \times 10 \times \frac{1}{2} = 50$ N

30. 00001.83

Sol. The measured value = 1.8 cm + 8 MSD $-$ 7 VSD

= 1.8 cm + 0.8 cm $-$ 0.77 cm

= 1.83 cm

Chemistry

PART – B

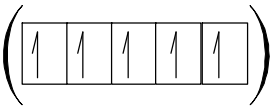
SECTION – A

31. B

Sol. $M - 4s^2 3d^6$

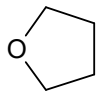
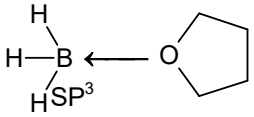
$M^{y+} - M^{+}\ 4s^1 3d^6$ – No. of unpaired electron is 5.

$x > y$

$M^{x+} - M^{+3}\ 4s^0\ 3d^5$ –No. of unpaired electron is 5  half-filled orbitals, more stable.

32. C

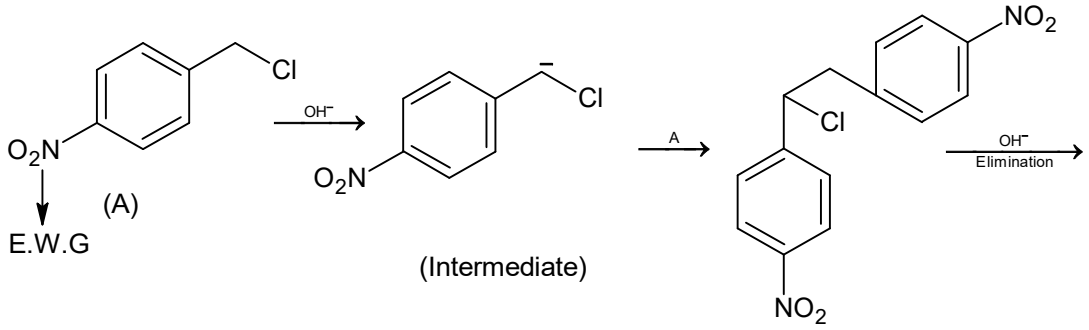
Sol. (I) Formation of $Na[Al(OH)_4]$

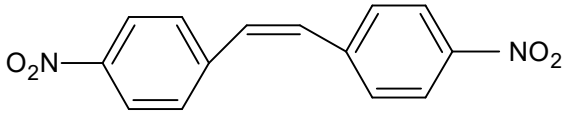
(II) $B_2H_6 +$  

(III) $SiF_4 + HF \longrightarrow H_2SiF_6$ (sp^3, sp^3d^2)

(IV) $PCl_5 \longrightarrow [PCl_4]^+[PCl_6]^-$ (sp^3d, sp^3, sp^3d^2)

33. D

Sol. 



Stilbene

34. C

Sol. 1 L weighs at NTP = 1.25 g

22.4 L at NTP will weigh $= 1.25 \times 22.4$ g

$= 28$ g

So, the gas can be N_2.

35. A

Sol. Specific conductivity of solution decreases with dilution due to decrease in concentration of electrolyte while λ_m increases due to (1) α increases (2). Interactions between ions decreases.

36. B

Sol. $P_T = P_A^o x_A + P_B^o x_B = P_A^o x_A + P_B^o (1 - x_A)$

$= P_A^o x_A + P_B^o - P_B^o x_A$

$= (P_A^o - P_B^o) x_A + P_B^o$

$P_A^o - P_B^o = 150,\ P_B^o = 50$

$P_A^o = 200,\ P_A^o / P_B^o = 200 / 50 = 4$

37. B

Sol. Freezing point and melting point is same for a pure substance

i.e $T_A = T_B$.

At freezing point entropy increases rapidly.

ΔS = +ve favourable factor.

ΔG = –ve spontaneous.

ΔU = –ve favourable process.

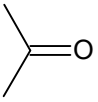
38. C

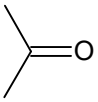
Sol. $SnO_2 + NaOH \longrightarrow Na_2SnO_3$

$[SnO_2]SnO_3^{-2} : 2Na^+$

39. B

Sol. In B lone pair of N undergo resonance

In A lone pair of N delocalized towards 

In C and D,  which is electron withdrawing.

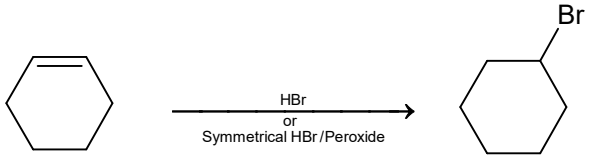
40. D

Sol. Molar mass is same for all. As branch/es increases surface area decreases, Van der Wall's force and boiling point decreases.

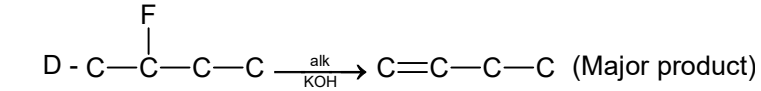
41. B

Sol. A – B.D.E. of alkyne is high

B – $Be_2C + H_2O \rightarrow Be(OH)_2 + CH_4$



C - Symmetrical alkene

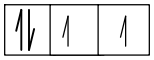
D - 

(Major product)

42. A

Sol. $\underset{3d^3}{Cr^{+3}} + NO \longrightarrow \underset{3d^4}{Cr^{+2}} + NO^+$

In strong ligand field  eg.

 t_{2g}.

No. of unpaired electron = 2

$\mu_s = \sqrt{2(2+2)} = \sqrt{8}\,BM$

43. B

Sol. $K_1 = K_2$

$\Rightarrow \frac{2}{3}$rd A has reacted for [A] = [C] = [D]

$K_1 + K_2 = \frac{1}{t}\ln\frac{[A_0]}{[A_0]/3}$, $t = \frac{1}{K_1+K_2}\ln 3$, $t = \frac{1}{2K_1}\ln 3 = \frac{1}{2K_2}\ln 3$

44. A

Sol. $AlBr_3 + NaOH \longrightarrow \underset{\text{white gelatineous ppt.}}{Al(OH)_3} + NaOH$

$\downarrow NaOH$

$\underset{\text{Soluble}}{NaAlO_2} + H_2O$

$AgNO_3 + AlBr_3 \longrightarrow \underset{\text{light yellow ppt}}{AgBr} + Al(NO_3)_3$

45. B

Sol. Cerussite $\rightarrow PbCO_3$, Siderite $\rightarrow FeCO_3$, Crocoite $\rightarrow PbCrO_4$, Calamine $\rightarrow ZnCO_3$.

46. C

47. C

Sol. Stainless steel – Fe, Cr, Ni
Solder – Sn, Pb
Bell metal – Cu and Sn
Brass – Cu and Zn

48. B

49. C

Sol. Acrolein is $CH_2 = CH - CHO$

50. C

Sol. Testosterone is a harmone
Ascorbic acid – Vitamin C
Thiamine – Vitamin B_1
Riboflavin – Vitamin B_2

SECTION – B

51. 00009.00

Sol.

Be	B	C	Al	Si	P	S	Sn	Pb
↓	↓	↓	↓	↓	↓	↓	↓	↓
Na_2BeO_2	Na_3BO_3	Na_2CO_3	$NaAlO_2$	Na_2SiO_3	NaH_2PO_2	$Na_2S_2O_3$	Na_2SnO_3	Na_2PbO_3
$+H_2$	$+H_2$	$+Na$ $+H_2$	$+H_2$	$+H_2$	$+PH_3$	$+Na_2S$	$+H_2$	$+H_2$

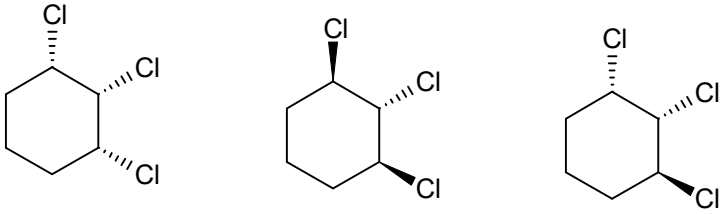
$$AlCl_3 + NaOH \longrightarrow \underset{\text{white gelatineous}}{Al(OH)_3} + NaCl$$

$$FeCl_3 + NaOH \longrightarrow \underset{\text{Reddish brown}}{Fe(OH)_3} + NaCl$$

$$SnCl_2 + NaOH \longrightarrow \underset{\text{White}}{Sn(OH)_2} + NaCl$$

52. 00003.00

Sol. 1, 2, 3-trichlorocyclohexane



3 geometrical isomers.

53. 00009.25

Sol. Meq. of $NH_4OH = 50 \times 0.1 = 5$ meq.

Meq. of $HCl = 50 \times 0.05 = 2.5$ meq.

	NH_4OH +	HCl	$\longrightarrow NH_4Cl + H_2O$
Initial	5meq	2.5 meq	0
After reaction	$(5-2.5)$	0	2.5 meq.
	$= 2.5$ meq.		salt

After reaction 2.5 meq. salt forms which is a salt of strong acid and weak base.
In the solution 2.5 meq. weak base and 2.5 meq. salt of same base is present, which is a buffer solution.

$$pOH = pK_b + \log\frac{[Salt]}{[Base]}$$

$$= 4.7447 + \log\frac{2.5}{2.5} = 4.7447$$

$$\therefore pH = 14 - 4.7447 = 9.2553$$

54. 00016.75

Sol. Vol. of N_2 = 20 ml

Pressure of dry N_2 = 725 – 25 = 700 mm

T = 300 K

$$PV = \frac{W}{M}RT$$

$$\frac{700}{760}\times\frac{20}{1000}=\frac{W}{28}\times 0.0821\times 300$$

$$W_{N_2}=\frac{7\times 2\times 28}{760\times 0.0821\times 300}=0.02094$$

$$\therefore \%\ \text{of}\ N_2=\frac{0.02094}{0.125}\times 100=16.752$$

55. 00004.00

Sol. $4\times 6\times 10^{23}$ atoms = 24×10^{23} atoms

24×10^{23} atoms of hydrogen – 24×10^{23} protons

$\Rightarrow 24\times 10^{23}$ electrons.

56. 00099.93

Sol. Vol. occupied by 1 mole of H_2O (g) at 373 k = nRT / P = $\frac{1\times 0.0821\times 373}{1}=30.62$ lt

Vol. of 1 mole of H_2O (l)

$=\frac{M}{\rho}=\frac{18}{0.958}=18.789$ ml $= 18.789\times 10^{-3}$ lt

% of vol. occupied by liquid water = $\frac{18.79\times 10^{-3}}{30.62}\times 100=0.0613\%$

% of free volume = 100 – 0.0613

= 99.9386%

57. 00069.06

Sol. $nH_2C=CH_2+n(CH_2=CHCl)\rightarrow [CH_2–CH_2–CH_2–CHCl]_n$

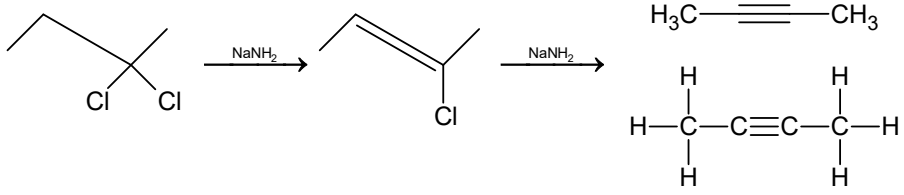
Mol. mass of C_2H_4 = 28, M. M of C_2H_3Cl = 62.5

$\therefore$ Empirical formula of copolymer = 28 + 62.5 = 90.5

% of vinyl chloride = $\frac{62.5}{90.5}\times 100$ = 69.06%

58. 00003.00

Sol.



48 + 6 = 54

$\left(\frac{54}{1+2}\right)\div 6$

$=\frac{18}{6}=3.00$

59. 00088.00

Sol. 1 mole CH_3CHO produces 1 mole Cu_2O

M. M of CH_3CHO = 44

M. M of Cu_2O = 143

$\therefore$ Produced $Cu_2O=\frac{2.86}{143}=\frac{2}{100}=0.02$ mole

$\therefore$ Mass of CH_3CHO in mixture = $0.02\times 44=0.88$

$\therefore$ % of CH_3CHO in mixture = $\frac{0.88}{1} \times 100 = 88\%$

60. 00003.87

Sol. $\underset{(B)}{\left[Cr(NH_3)_4Cl_2\right]Br} \xrightarrow{AgNO_3} \left[Cr(NH_3)_4Cl_2\right]^+ + NO_3^{-1} + \underset{\substack{\text{Yellow precipitate} \\ \text{Partially soluble in concentrated } NH_3 \text{ i.e } \left[Ag(NH_3)_2\right]Br}}{AgBr}$

$\underset{(A)}{\left[Cr(NH_3)_4Br.Cl\right]Cl} \xrightarrow{AgNO_3} \left[Cr(NH_3)_4ClBr\right] + NO_3^- + \underset{\text{White ppt}}{AgCl}$

Soluble in concentrated NH_3, i.e. $\left[(AgNH_3)_2\right]Cl$

Hybridisation of Cr in A or B $\rightarrow d^2sp^3$ having 3 unpaired electrons.

$\mu = \sqrt{3(3+2)} = \sqrt{15}$ = 3.8729 = 3.87

Mathematics

PART – C

SECTION – A

61. C

Sol. $T_r = \frac{2r^2+7r+4}{r(r+1)(r+2)2^r} = \frac{2(2r+1)(r+2)-(2r+3)r}{r(r+1)(r+2)2^r} = \frac{2r+1}{r(r+1)2^{r-1}} - \frac{2r+3}{(r+1)(r+2)2^r}$

$\therefore S_n = V_0 - V_n = \frac{3}{1.2} - \frac{2n+3}{(n+1)(n+2)2^n}$

$\therefore S_\infty = \frac{3}{2}$

62. A

Sol. $N = \frac{20!}{(5!)^4\,4!}4!(4!)^4 = \frac{20!}{5^4}$

63. D

Sol. Favourable cases

{1, 2, 3, 4, ….., 10, 20, 30, 40, ….., 100, 200, 300, ….., 900, 1000}

$P(E) = \frac{28}{1000} = \frac{7}{250}$

64. C

Sol. $\frac{2!}{(n+1)(n+2)}\sum_{r=0}^{n}(-2)^r\ {}^{n+2}C_{r+2}$

$= \frac{2!}{(n+1)(n+2)}\left[{}^{n+2}C_2 - {}^{n+2}C_3(2) + {}^{n+2}C_4(2)^2 + ... {}^{n+2}C_{n+2}(-2)^n\right]$

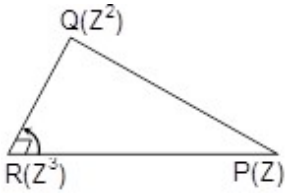
$= \frac{2!}{4(n+1)(n+2)}\left[(1-2)^{n+2} - 1 + 2(n+2)\right] = \frac{1}{n+2}$; n is odd

$= \frac{1}{n+1}$; n is even

65. D

Sol. $\text{Arg}\left(\frac{Z^2-Z^3}{Z-Z^3}\right) = \frac{\pi}{2} \Rightarrow \text{Arg}\left(\frac{Z(1-Z)}{(1-Z)(1+Z)}\right) = \frac{\pi}{2}$

$\text{Arg}\left(\frac{Z}{1+Z}\right) = \frac{\pi}{2}$; $Z \neq 0, -1 \Rightarrow \frac{Z}{1+Z}$ is purely imaginary



66. B

Sol. $I_1 = \int_0^{\pi/3}(2\sec x)^{100}\,dx$

Let sec x + tan x = e^t

$\therefore I_1 = \int_0^{\ln(2+\sqrt{3})} 2\left(e^t + e^{-t}\right)^{99} dt \Rightarrow I_1 = 2I_2$

67. B

Sol. Use g(f(x)) = x

68. C

Sol. $\int_1^5 \left(\left(\sqrt{x-1}+1\right)+\left|\sqrt{x-1}-1\right|\right)dx$

$= \left(\frac{2}{3}(x-1)^{\frac{3}{2}}+x\right)_1^5 + \int_1^2 \left(1-\sqrt{x-1}\right)dx + \int_2^5 \left(\sqrt{x-1}-1\right)dx$

$= \left(\frac{2}{3}(x-1)^{\frac{3}{2}}+x\right)_1^5 + \left(x-\frac{2}{3}(x-1)^{\frac{3}{2}}\right)_1^2 + \left(\frac{2}{3}(x-1)^{\frac{3}{2}}-x\right)_2^5$

$= \frac{16}{3}+5-1+1-\frac{2}{3}+\frac{16}{3}-5-\left(\frac{2}{3}-2\right) = \frac{34}{3}$

69. C

Sol. $(x-2)(y-1)=4$

$XY = 4$

If normal at $\left(ct, \frac{c}{t}\right)$ intersects curve again at $\left(ct', \frac{c}{t'}\right)$, then $t' = -\frac{1}{t^3}$

Now, $2t = 4 \Rightarrow t = 2 \therefore\ t' = -\frac{1}{8}$

$X' = -\frac{1}{4}$, $Y' = -16$

$\therefore\ \alpha = -\frac{1}{4}+2$, $\beta = -16+1$

$\alpha = \frac{7}{4}$, $\beta = -15$

70. B

Sol. $f(x) = \frac{5}{2}\cos^2\pi x - \sin 3\pi x$; $x \in [0, 1]$

$f(t) = \frac{5}{2}-\frac{5}{2}t^2-3t+4t^3$; $t \in [0, 1]$

$f'(t) = 12t^2 - 5t - 3 = 0 \Rightarrow t = \frac{5\pm 13}{24} = -\frac{1}{3}, \frac{3}{4}$

$f(0) = \frac{5}{2}$, $f\left(\frac{3}{4}\right) = 4\times\frac{27}{64}-\frac{5}{2}\times\frac{9}{16}-3\times\frac{3}{4}+\frac{5}{2} = \frac{108-90-144+160}{64} = \frac{17}{32}$

$f(1) = 4-\frac{5}{2}-3+\frac{5}{2} = 1$

71. A

Sol. $\lim_{y\to 0^+} \frac{e^y\left(y^3+1\right)-1}{y^2}$; $\left(\frac{0}{0}\right)$ form

72. C

Sol. $\frac{ar_1+br_2+cr_3}{\Delta} = r_1\left(\frac{1}{r_2}+\frac{1}{r_3}\right)+r_2\left(\frac{1}{r_1}+\frac{1}{r_3}\right)+r_3\left(\frac{1}{r_1}+\frac{1}{r_2}\right) = \frac{4R}{r}-2 = 6$

$\Rightarrow \frac{R}{r} = 2$

73. B

Sol. $A \equiv \left\{0, \frac{2\pi}{5}, \frac{\pi}{2}, \frac{4\pi}{5}\right\}$

74. A

Sol. Shown in the table

p	q	~q	$p \to \sim q$	$q \to p$	$(p \to \sim q) \vee (q \to p)$
T	T	F	F	T	T
T	F	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T

75. C

Sol. $\text{var}(\lambda x_i) = \lambda^2 \text{var}(x_i')$

76. A

Sol. The equation of the plane through the line of intersection of the planes is

$4x + 7y + 4z + 81 + \lambda(5x + 3y + 10z - 25) = 0$

$\because (4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + 81 - 25\lambda = 0$

$\therefore \frac{4+5\lambda}{1} = \frac{7+3\lambda}{-4} = \frac{4+10\lambda}{6}$

$\Rightarrow \lambda = -1$

$\therefore$ Required plane is $x - 4y + 6z = 106$

77. C

Sol. $xf(x) + \int_0^x f(t)dt = 8x^3 - 9x^2 + 12x$

$\Rightarrow xf'(x) + 2f(x) = 24x^2 - 18x + 12$

$\Rightarrow f'(x) + \frac{2}{x}f(x) = 24x - 18 + \frac{12}{x}$ $\therefore$ Linear differential equation in y

78. A

Sol. $f(x) = ax^5 + bx^4$

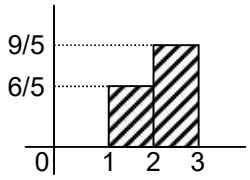
$\lim_{x\to 0} \frac{f(-x)}{2x^4} = 1 \Rightarrow \lim_{x\to 0} \frac{-ax + b}{2} = 1 \Rightarrow b = 2$

Also, $e^{\lim_{x\to 0} \frac{1}{x}\left(\frac{-ax+b}{2} - 1\right)} = e^{-4} \Rightarrow a = 8$

$\therefore f(x) = 8x^5 + 2x^4$

79. D

Sol. Required area = $\frac{6}{5} + \frac{9}{5} = 3$



80. B

Sol. PA = PB

$\Rightarrow$ P lies on the perpendicular bisector of line segment joining A and B

SECTION – B

81. 00054.00

Sol. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ and $ad - b^2 \neq 0$

b = 0, then number of matrices are 9

b = 1, 2, 3, then number of matrices are 15 each

82. 00333.00

Sol. f(1) = a + b + c = 0 and 102 < 36a + 6b + c < 110

$\Rightarrow \frac{102}{5} < 7a + b < 22 \Rightarrow 7a + b = 21$ and 196 < 64a + 8b + c < 206

$\Rightarrow 28 < 9a + b < \frac{206}{7} \Rightarrow 9a + b = 29 \Rightarrow a = 4, b = -7, c = 3$

$\therefore$ f(10) = 100a + 10b + c = 333

83. 00016.00

Sol. (i) $x^2 - 8x + 6 = 1$

(ii) $x^2 - 9 = 0$

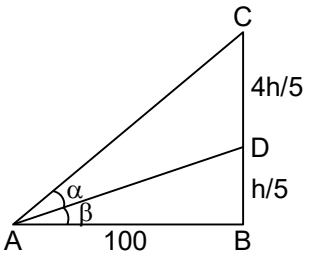
(iii) $x^2 - 8x + 6 = -1$

84. 00500.00

Sol. $\tan(\alpha + \beta) = \frac{h}{100}$

$\tan\alpha = \frac{2}{3}$

$\tan\beta = \frac{h}{500} \Rightarrow \frac{\frac{2}{3} + \frac{h}{500}}{1 - \frac{2}{3}\frac{h}{500}} = \frac{h}{100}$



$\Rightarrow$ h = 100, 500 ($\because \tan\beta > 1$)

$\therefore$ h = 500

85. 00108.00

Sol. $\vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow |\vec{a} + \vec{b}| = |\vec{c}|$

$\Rightarrow 36 + |\vec{b}|^2 + 60 = 196 \Rightarrow |\vec{b}| = 10$

$|\vec{a} \times \vec{b} + \vec{b} \times \vec{c}| = |\vec{b} \times (\vec{c} - \vec{a})| = |(\vec{a} + \vec{c}) \times (\vec{c} - \vec{a})| = 2|\vec{a} \times \vec{c}|$

$|\vec{a} + \vec{c}| = |\vec{b}| \Rightarrow 36 + 196 + 2(\vec{a} \cdot \vec{c}) = 100 \Rightarrow \vec{a} \cdot \vec{c} = -66$

$\therefore |\vec{a}||\vec{c}|\cos\theta = -66 \Rightarrow \cos\theta = -\frac{11}{14}$

$|\vec{a} \times \vec{b} + \vec{b} \times \vec{c}|^2 = 4|\vec{a} \times \vec{c}|^2 = 4 \times 36 \times 196 \times \frac{75}{196} = 10800$

86. 00007.50

Sol. $S \equiv (x-1)^2 + (y-1)^2 - 4 = 0$

Equation of normal to $(y + 11)^2 = 2(x - 4)$

$y + 11 = -t(x - 4) + t + \frac{t^3}{2}$. It passes through (1, 1)

$\Rightarrow t^3 + 8t - 24 = 0 \Rightarrow t = 2 \therefore P(6, -9) \therefore \Delta OCP = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 6 & -9 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 7.5$

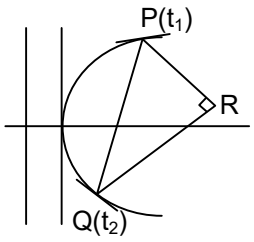
87. 02000.00

Sol. PQ is focal chord

$\therefore t_1t_2 = -1$

$a\left(t_1 + \frac{1}{t_1}\right)^2 = \frac{200}{9} \Rightarrow \left|t_1 + \frac{1}{t_1}\right| = \frac{10}{3}$

Area of $\Delta PQR = \frac{a^2}{2}\left|t_1 + \frac{1}{t_1}\right|^3 = 2 \times \left(\frac{10}{3}\right)^3 = \frac{2000}{27}$



88. 00064.00

Sol. $\frac{\pi}{2} - \cot^{-1}\cot\left(3 - \frac{3}{x^2 + 3|x| + 8}\right) = \frac{\pi}{2} + \frac{3}{9|x|} - 3$

$\Rightarrow x^2 + 3|x| + 8 = 9|x| \Rightarrow x^2 - 6|x| + 8 = 0 \Rightarrow x = \pm 2, \pm 4$

89. 00110.00

Sol. $\Delta = \begin{vmatrix} a & 3 \\ 4 & -3 \end{vmatrix} = 0 \Rightarrow -3a - 12 = 0 \Rightarrow a = -4$

$\Delta_x = \begin{vmatrix} \lambda & 3 \\ \mu & -3 \end{vmatrix} = -3(\lambda + \mu) \neq 0$

$\Delta_y = \begin{vmatrix} a & \lambda \\ 4 & \mu \end{vmatrix} = -4(\lambda + \mu) \neq 0$

90. 00300.00

Sol. $ab = 2^{18}\, 3^8\, 5^4\, 7^2\, 11^1.13^1.17^1.19^1$

If $\frac{a}{b} \in$ integer then number of ordered pairs (a, b) = $10 \times 5 \times 3 \times 2 = 300$